

Demonstration of Proposed Features

Date	05 July 2026
Team ID	SWTID-2026-5245
Project Name	OptiCrop – Smart Agricultural Production Optimization Engine
Maximum Marks	1 Mark

Demonstration of Proposed Features:

This document captures all the features that were proposed during the project planning phase and tracks whether each feature was successfully implemented and demonstrated. It serves as evidence of the team's ability to deliver on the proposed solution.

S.No	Feature Name	Description	Status (Implemented/Partial/Pending)	Demonstrated (Yes/No)	Remarks
1	Soil & Climate Input Form	Web form for the farmer to enter N, P, K, temperature, humidity, pH and rainfall.	Implemented	Yes	Working end-to-end on the live deployment
2	Data Preprocessing	Clean and preprocess the agricultural dataset	Implemented	Yes	Successfully tested
3	Crop Recommendation Engine	Trained scikit-learn classifier predicts the most suitable crop from submitted values.	Implemented	Yes	Core ML feature functioning correctly
4	Confidence Score Display	Shows the model's prediction confidence (%) alongside the recommended crop.	Implemented	Yes	Displayed on the result page
5	Client-Side Input Validation	Min/max/step constraints on each input field to catch invalid values before submission.	Implemented	Yes	Present on all 7 input fields
6	Web Deployment	Application deployed as a serverless Flask app on Vercel with a public URL.	Implemented	Yes	Live at https://opticrop-indol.vercel.app/

Feature Implementation Summary:

Total Features Proposed	6
Total Features Implemented	6
Total Features Demonstrated	6
Overall Implementation Rate (%)	100%